Inside Family Caregivers' Conversations with a Well-being Chatbot in the Wild: Intent, Affect, and Supportive Moves from an 8-Week Deployment

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Abstract

Family caregivers of people living with dementia face accumulating stressors, yet traditional mental health support is hard to reach amid unpredictable, demanding caregiving. Large language model (LLM)-based well-being chatbots have emerged as an on-demand stopgap: in our 8-week deployment, we observed medium-to-large improvements in caregivers' anxiety, depression, anger, perceived stress, and sleep. Yet how caregivers use such tools, and how the tools respond, remains underexplored. We analyzed the deployment's chat logs of 13,637 messages between 22 caregivers and the chatbot. Through an LLM-assisted workflow, we coded caregivers' communicative intent (speech act theory) and affect, and the chatbot's supportive moves (emotional-support and helping-skills frameworks). Caregivers used the chatbot mainly to express feelings and report situations, with variation across individuals; their conversations moved through three paths resembling coping processes. The chatbot enacted a consistent reflective style, mirroring emotion and asking questions, and left roughly a third of explicit guidance requests without advice in its immediate response. We contribute an empirical account of the caregiver-chatbot interaction and raise two questions for the community: how to meet caregivers where they are by making the chatbot's approach context-aware, and what to code, and at what unit, to tell whether its supportive moves land.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**.

Keywords

family caregivers; conversational AI; large language models; well-being support; chat log analysis

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1 Introduction

Family caregivers who support their loved ones living with dementia experience elevated levels of stress, depression, anxiety, and sleep disruptions that compromise their mental well-being [1]. Yet, the same demands that undermine caregivers' well-being, including unpredictable schedules and competing responsibilities, also limit their access to traditional support, such as in-person therapy and psychoeducation, which is time-intensive and requires advance planning [4, 13, 18]. Large language model (LLM)-based chatbots have emerged as a candidate stopgap, with on-demand availability that fits caregivers' schedules and dynamic responses to the diverse challenges they face [2, 16, 17]. Our recently completed 8-week field deployment with 23 dementia caregivers offers preliminary evidence of this promise: findings showed medium-to-large improvements pre- to post-deployment on self-report measures across anxiety, depression, anger, perceived stress, and sleep, and companion interviews surfaced positive but mixed experiences of how well the chatbot met caregivers' well-being needs. However, much less is known about what these interactions consist of over weeks of everyday use: what caregivers try to accomplish in talking with the chatbot, what emotions they express, how the chatbot responds, and how conversations unfold. To date, caregiver-facing chatbots have mostly been studied through interviews and surveys that capture caregivers' self-reported needs and preferences after short, study-based sessions with a prototype [5, 17], rather than analyzing the conversations themselves. Naturalistic, longitudinal logs from this vulnerable population are accordingly rare, and the methods to characterize them are still developing.

We analyze the conversation logs from this deployment; one of the 23 caregivers did not share their logs, so our analysis draws on the remaining 22. We characterize the caregiver side through communicative intent, grounded in speech act theory [7, 15] and paired with sentiment analysis, and the chatbot side through a move-coding scheme grounded in emotional-support dialogue and helping-skills frameworks [6, 9]. Both analyses are scaled with

LLM assistance via U-M GPT, the University of Michigan's secure institutional LLM service.

This work makes three contributions. First, we provide an empirical account of what caregivers do and how the chatbot responds across 8 weeks of real-world use, characterizing both sides of these naturalistic, longitudinal interactions that are rarely examined in this population. Second, we demonstrate an LLM-assisted approach for coding both sides of these support conversations on a secure institutional service. Third, we use this account to surface, for community discussion, two open questions for designing and evaluating conversational AI for caregivers' mental health responsibly: how to meet caregivers where they are by making the chatbot's approach context-aware; and what to code, and at what unit, to tell whether the chatbot's supportive moves land.

2 Methods

2.1 Study Dataset

The chat log data come from our 8-week deployment of an off-the-shelf LLM-based mental health chatbot with family caregivers of people living with dementia who experienced mild-to-moderate caregiver burden, as determined by the Zarit Burden Interview [3]. The chatbot, Earkick,[1] is designed around reflective listening and non-prescriptive emotional probing; it was introduced as a tool for participants' own well-being, with no prescribed usage frequency or topic. Of the 23 participants who completed the 8-week study, 22 shared their chat logs, which form our corpus: 463 conversation sessions and 13,637 messages, of which 3,147 were from caregivers and the remainder from the chatbot. The chatbot accounts for more messages because it often splits a single turn into several shorter messages. The study was approved by the University of Michigan Institutional Review Board (HUM00271412), and all LLM-assisted coding described below was run on U-M GPT, the University of Michigan's institutional LLM service.

2.2 Coding Caregiver Communicative Intent

We coded each caregiver message for its communicative *intent*. Following speech act theory [15], which classifies utterances by their illocutionary force, and Jakobson's phatic function [7] for contact-maintaining acts, we organized intent into five directions: Assertive, Expressive, Directive, Commissive, and Phatic (omitting declarations, which require institutional authority absent from this setting). Coding proceeded in three stages combining LLM assistance with researcher judgment. We first drew a 239-message sample using participant-stratified proportional allocation with a floor of three messages per participant, and had GPT-5.2 open-code a free-text intent label for each, grounded in the five directions and conditioned on the preceding context (the prior caregiver message and chatbot turns). The first author consolidated these labels into a codebook of 13 categories spanning the five directions (Table 1), with each message assigned a single primary intent. We then applied the codebook to the full corpus with GPT-4o; this reached 74.9% agreement with the author's labels on the 239-message sample.

[1] https://earkick.com/

2.3 Coding Chatbot Moves

To characterize the chatbot's turns, we adapted the emotional-support strategy scheme of Liu et al. [9], which itself builds on Hill's helping-skills taxonomy [6]. We tested Liu's eight strategies on a proportionally stratified 127-message sample, and they applied. Because several recurrent moves (opening/check-in, closing, acknowledgment/rapport, safety/redirect) would otherwise collapse into Liu's residual Others category, we broke them out as dedicated codes, yielding 12 categories that retain all of Liu's strategies (Table 2). We coded one primary move per message, defined as the move carrying the message's main supportive work. We then applied the codebook to the full corpus with an LLM, using category definitions and boundary rules. On a held-out set of 100 messages, model labels agreed with the author's labels at 85.0% (Wilson 95% CI [76.7, 90.7]; Cohen's $\kappa = 0.816$). The most frequent residual confusion was between restatement and reflection of feelings, which the chatbot often bundles in a single message.

2.4 Sentiment Analysis

We ran two off-the-shelf RoBERTa classifiers on every caregiver message: a sentiment classifier[2] to obtain continuous valence as $P(\text{positive}) - P(\text{negative})$, a score in $[-1, +1]$; and a GoEmotions classifier[3] trained on the 27 emotion categories to obtain discrete emotion labels, which we use descriptively to name what a valence shift is composed of. Both classifiers were used as published.

3 Findings

We organize the findings around what caregivers' messages do (§3.1), how the chatbot responded (§3.2), how individual conversations unfolded (§3.3), and how affect shifted within them (§3.4).

3.1 What Caregivers' Messages Do

Across the 8-week deployment, caregivers' messages were dominated by Expressive and Assertive acts, each occupying roughly 30-40% of messages, with Phatic acts around 15-22%, Directive acts low (about 2-10%), and Commissive acts a thin remainder (Figure 1a). Caregivers were predominantly narrating situations, expressing feelings, and reflecting, rather than requesting guidance. This pattern likely reflects two co-occurring forces: how caregivers preferred to use the space, and what the chatbot's style encouraged (Earkick's reflective stance tends to invite continued narration and disclosure rather than question-and-answer exchange). This aggregate mix held roughly steady across the 8 weeks, with only a modest shift observed. However, individuals differed in how they engaged the chatbot. Caregivers' intent profiles fell into several recognizable patterns: some used it largely for Expressive acts, others leaned Directive or Assertive, a few were more Phatic or balanced (Figure 1b).

3.2 How the Chatbot Responds

The chatbot's support moves were consistent with a reflective, or active, listening stance (Figure 2). Its most common responses were asking a question that invited the caregiver to say more (`Question`)

[2] https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest
[3] https://huggingface.co/SamLowe/roberta-base-go_emotions

Table 1: Caregiver intent codebook: 13 categories across five communicative directions, grounded in speech act theory [15].

Direction	Category	Definition
Assertive	Daily status/situational reporting	Reports what is happening, what happened, or what is planned (e.g., events, activities, routines, or states).
	Explaining/reasoning	Provides a cause, hypothesis, or mechanism to explain the external situation or another person's behavior.
	Stating/evaluating coping strategies	Identifies or evaluates a strategy or approach for managing the caregiving situation, emotions, or well-being.
Expressive	Emotional state disclosure	Briefly names or labels a current emotional state, without extended narrative.
	Venting/narrating distress	Recounts stressful events with the primary purpose of offloading emotional weight.
	Reflecting/meaning-making	Interprets the caregiver's own relationship to an experience (e.g., meaning, identity, values, or growth).
	Ambivalence/moral conflict	Articulates conflicted feelings or moral tension (e.g., guilt, push-pull about decisions, competing values).
	Positive affect/gratitude	Shares positive emotion (e.g., joy, relief, hope, gratitude, or spiritual comfort) rather than reporting a state.
Directive	Requesting practical guidance	Asks for concrete advice, information, instructions, options, or strategies.
	Managing chatbot interaction	Instructs the chatbot on communication style, sets behavioral boundaries, grants/withholds consent, or signals (dis)engagement.
Phatic	Acknowledging/affirming	Signals agreement, validation, appreciation, or receipt without adding substantive content.
	Opening/closing the channel	Greets, introduces, signs off, or signals unavailability to continue.
Commissive	Committing to future action	Pledges a specific future action (e.g., trying a suggestion, implementing a plan, returning later).

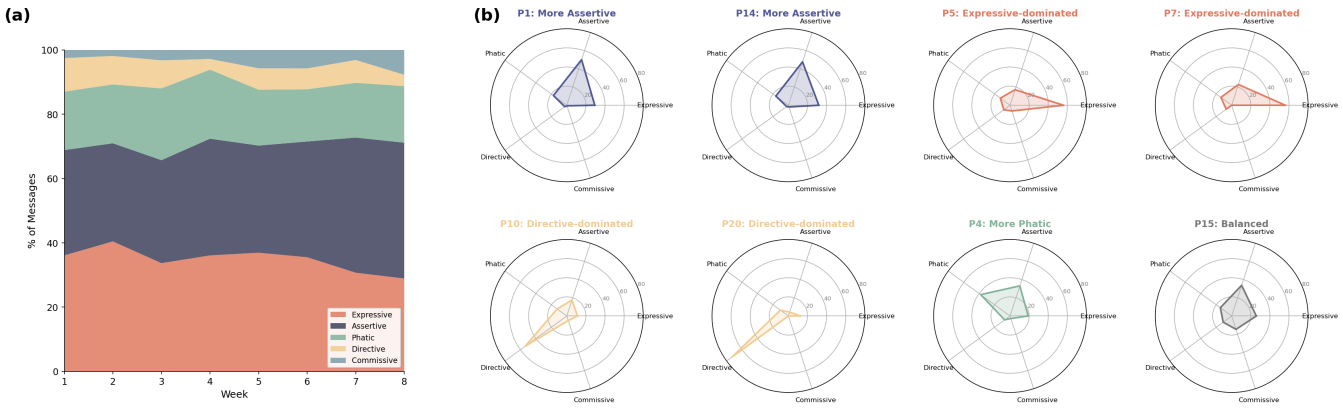


Figure 1: Caregiver communicative intent across the deployment. (a) Weekly share of messages by intent act, aggregated across caregivers. (b) Per-caregiver intent profiles for eight representative participants.

and naming the emotion the caregiver had expressed (`Reflection of feelings`), followed by validating or reassuring the caregiver (`Affirmation`) and paraphrasing the content of what they had said (`Restatement`). Together these make up roughly three-quarters of all chatbot messages. In contrast, concrete problem-solving was uncommon: offering advice (`Providing suggestions`) and sharing factual information (`Information`) together account for under 8% of chatbot messages.

Two patterns in how the chatbot ordered these moves within a single turn also emerge, where a turn is the run of consecutive messages the chatbot sent before the caregiver replied. First, `Reflection of feelings`, `Affirmation`, and `Restatement` formed a tightly connected cluster, transitioning among one another more often than to any other move. Second, a binomial test on move position shows that this cluster appeared above chance in a turn's opening messages, and `Question` and `Providing suggestions` above chance in its closing messages. The characteristic turn thus opens by reflecting the caregiver's utterance back and closes by handing the floor back with a question.

Within each turn, the chatbot's opening support move was contingent on the caregiver's communicative intent ($\chi^2(132) = 1711.4$, $p < .001$, Cramér's $V = .23$), a pattern concentrated in a few strongly coupled intent–move pairs.[4] Across four of the caregivers'

[4] p-values were Benjamini-Hochberg corrected for multiple comparisons; all reported residuals are significant at $q < .001$.

Table 2: Chatbot move codebook: 12 categories, grounded in emotional-support dialogue and helping-skills frameworks [6, 9].

Move	Description
Question	Invites the caregiver to share, elaborate on, or explore their situation, thoughts, or feelings.
Reflection of feelings	Names or mirrors back the caregiver's emotion, including one implied but not stated.
Restatement/paraphrasing	Rephrases or mirrors the content of what the caregiver said.
Affirmation/reassurance	Affirms strengths or efforts, validates or normalizes a reaction, or reassures.
Providing suggestions	Offers a concrete, usable coping strategy, action, or next step.
Information	Provides facts, resources, psychoeducation, or an explanation.
Self-disclosure	Expresses, in the first person, a simulated reaction or shared experience.
Acknowledgment/rapport	Minimal receipt or relational move; brief acknowledgment or banter.
Opening/check-in	A session-initiating greeting that invites the caregiver to begin.
Closing	Signs off, wishes the caregiver well, or wraps up the session.
Safety / redirect	Flags risk or crisis and urges professional or emergency help.
Miscellaneous	Primary function fits none of the above (e.g., app logistics).

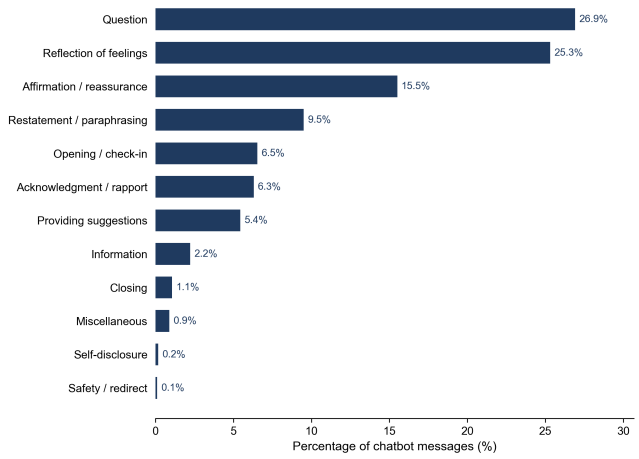


Figure 2: Distribution of chatbot moves (N = 10,490 messages).

Expressive intents (i.e., venting, emotional state disclosure, ambivalence and moral conflict, and reflecting and meaning-making), the chatbot's first move was predominantly `Reflection of feelings` (56-71%; adjusted residuals +4.1 to +9.8), returning the expressed emotion to the caregiver. When caregivers instead committed to a future action, the dominant opening move shifted to `Affirmation` (71%; adjusted residual +12.3), validating the stated commitment. Directive requests for practical guidance, by contrast, elicited no single dominant opening move. Widening the lens to the full chatbot turn, 69% of multi-message turns following a guidance request contained at least one advice-bearing move (i.e., `Providing suggestions` and `Information`). Yet this also means that nearly one-third of these turns contained no advice-bearing move at all.

3.3 How Conversations Unfold

We first asked where in a session each intent tends to appear. In sessions with at least three caregiver messages, we tagged each message with a within-session position (i.e., opening, middle, or closing) by its relative ordinal location, then compared each intent category's share at each position against its corpus-wide share. We found openings over-represented daily status reporting (+17.6%) and emotional state disclosure (+9.5%), implying that caregivers are more likely to begin by setting the scene, reporting what is happening and how they feel. Requesting practical guidance is also over-represented at openings (+8.7%), suggesting that when caregivers do come with a question, they tend to come with it up front. Middles carried deeper processing intent, over-representing reflecting or meaning-making (+2.4%), which was noticeably rare at openings (-9.3%). Closings shift toward conversational maintenance: acknowledging (+8.6%), opening/closing acts (+7.4%), and committing to future action (+3.6%). Deeper-processing intents (i.e., reflecting/meaning-making, explaining/reasoning, and stating/evaluating coping) climbed from 7.7% of messages in single-message sessions to 33.6% in sessions of eight or more, rising with session length.

We next examined whether caregivers' communicative intents unfold in recurring sequences. We collapsed runs of consecutive same-intent messages so that transitions capture movement between different communicative intents. For each ordered intent pair, we compared how often one intent followed another to that intent's overall frequency in the collapsed sequence. The fifteen most over-represented transitions organize into three recurring paths (Figure 3).[5] **(a) A reflective processing loop** moved from daily status reporting through reasoning about the situation and evaluating coping strategies into reflective meaning-making, with reflecting sometimes looping back to explaining. This suggests caregivers oscillate between interpreting personal meaning and re-examining the external situation that produced it. **(b) Expression into contextualization**: after an emotional disclosure, caregivers were likely either to ground the feeling in a concrete event or to move into deeper interpretation, and ambivalence similarly led to reflecting or explaining. **(c) A short task-oriented exchange**: caregivers' requests for practical guidance are followed by acknowledgment, likely depicting caregivers coming with a question, receiving an answer, and signaling receipt.

3.4 What Happens to Affect

Finally, we examined within-session affective movement. For sessions of at least five caregiver messages, we partitioned messages into opening, middle, and closing thirds and stratified sessions by the mean valence of their opening third (Figure 4a). Sessions that began negative moved toward neutral over their course; sessions that began positive drifted mildly downward toward neutral; sessions that began neutral stayed neutral. For sessions that began negative, the discrete emotions show that this movement was a recomposition rather than a simple fading of affect (Figure 4b). The opening carried acute negatives, sadness most of all, alongside

[5]These paths are the caregiver-side projection of a dyadic exchange, in which each caregiver message follows a chatbot reply; future work should examine how chatbot replies shape these paths.

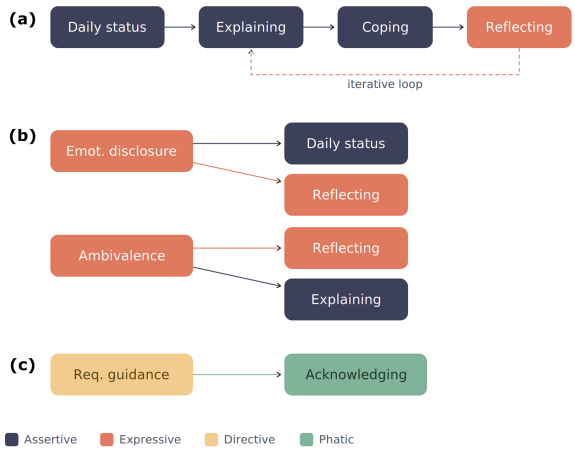


Figure 3: Three recurring within-session transition paths among caregiver communicative intents: (a) a reflective processing loop, (b) emotional expression into contextualization, and (c) a short task-oriented exchange.

disappointment and annoyance; by the middle these had softened, sadness falling sharply, with approval rising to the front of the remaining emotions; and the closing consolidated into approval, admiration, and gratitude. We are cautious about reading a therapeutic mechanism into this shift: stratifying on a negative opening invites regression toward the mean, and closing turns carry their own positive register of acknowledgments and farewells. We therefore report it as a descriptive regularity and defer to the discussion what measuring in-session change would require.

4 Discussion

In this section, we read findings for what they imply about designing and evaluating a chatbot that meets caregivers where they are.

The three within-session paths can be read against the literature on stress and coping, which distinguishes problem-focused, emotion-focused, and meaning-focused coping [8, 11]. In the stress-process model, these coping approaches mediate between caregiving stressors and caregivers' well-being [12]. The reflective processing loop resembles meaning-focused coping; the path from emotional disclosure into contextualization sits between emotion regulation and meaning-making; and the request-to-acknowledgment exchange resembles problem-focused coping. When conversations followed these paths, caregivers' messages were doing the kinds of communicative work this literature associates with adjustment. This offers a conversation-level view of what those medium-to-large improvements in self-reported outcomes looked like from inside the 8 weeks caregivers spent with the chatbot, though not of how those outcomes were produced.

The chatbot side of these conversations had a consistent character. It leaned on questions, reflection, and affirmation and rarely advised. This pattern aligns with active-listening practice [14] and with the core skills of motivational interviewing [10], in which withholding immediate advice is a deliberate move rather than a failure to help. This approach fits expressive and reflective intents well, but sits less aligned with explicit requests for guidance: as §3.2 shows, nearly a third of turns following the caregiver's guidance request carried no advice-bearing move. This can be viewed as a gap in responsiveness, or as faithful reflective practice, and how to read this depends on what a given caregiver needed at that moment. Two considerations follow. The first concerns the caregiver's momentary availability. Findings showed that caregivers' messages with reflecting/meaning-making intent unfold mostly in longer exchanges (§3.3), yet caregiving is fragmented and cognitively taxing [13]. The moments when a caregiver most needs to work through a difficult feeling may be the moments they can least sustain a multi-turn conversation. The most therapeutically promising path the chatbot offers may thus be the one caregivers are least free to follow when they need it. Reading that availability from signals such as time of day, location, input modality, and message length is a context-inference problem of the kind ubiquitous computing is equipped to address. A system attuned to these cues could compress reflective work when a caregiver's bandwidth is low, or ensure a concrete answer arrives when someone is plainly rushed. The second is a difference between caregivers rather than within a session: caregivers differed in the intents they brought, with some noticeably more directive than others (§3.1). If such differences prove stable enough to act on, this raises the possibility that a structured, advice-forward style serves some caregivers while a purely reflective one serves others. Whether and how the reflect-advise balance should adapt is a question we leave open.

Moving toward answers to questions like these would take evaluation a grain finer, down to what happens from one message to the next (e.g., whether a caregiver has the bandwidth to take up a reflective prompt, whether a `Question` actually draws them into the exchange it opens). The move coding we demonstrate is suited to one kind of question. It supports a fidelity check, telling us whether the chatbot behaved consistently with its intended reflective design, which in our deployment it did. What it does not tell us is whether those moves were taken up (e.g., whether a reflective prompt actually drew the caregiver into reflection, or whether a question drew out the elaboration it invited). Learning which moves land, and which land in the ways we would therapeutically hope to see, is what would let us shape how such a chatbot behaves, and it turns on finding the right unit at which to read that correspondence. The caregiver side poses a parallel question of what to code in the first place. The current communicative intent captures what a message does, not whether the caregiver is doing something we would want to encourage in its content (e.g., reframing a distressing thought toward a more balanced appraisal), which would mean coding for a different target. Pursuing that finer grain is not free, and raises additional practical questions for the field to take up: whether an LLM-assisted pipeline stays reliable enough to scale to subtler coding, and how to read message-to-message correspondence in the grain of real deployment, where a chatbot turn splits across several messages, and a caregiver may take up a prompt several turns later, or not at all. Together, these mark out where message-level evaluation of such systems will have to develop, and we bring them to this community as questions worth working through together.

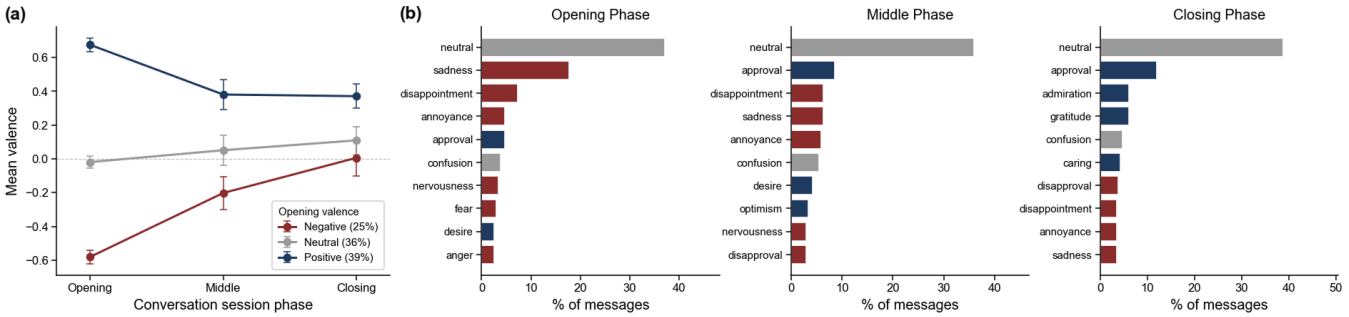


Figure 4: (a) Within-session valence by opening valence; (b) top emotions at the opening, middle, and closing phase of negatively-starting sessions.

5 Conclusion

We analyzed 13,637 messages from an 8-week deployment of an LLM-based mental health chatbot with 22 dementia caregivers, characterizing caregivers' communicative intent and affect alongside the chatbot's support moves. Across caregivers, messages more often expressed feelings and reported situations than requested guidance, though individuals varied in how they used the chatbot. The chatbot enacted a consistent reflective style, favoring reflection, affirmation, and restatement over advice; this aligned with caregivers' predominant expressive intents, yet nearly a third of explicit guidance requests drew no advice in its immediate response. Within sessions, caregivers' messages moved through three recurring intent paths that we read against emotion-, meaning-, and problem-focused coping; and in sessions that began negative, valence moved toward neutral as the mix of expressed emotions shifted. Based on these findings, we raise two questions. The first is a design question. The chatbot's minimal-guidance style leaves more processing to caregivers, but reflective intent is concentrated in longer sessions that caregivers cannot always sustain. How should a chatbot balance reflective and guidance support, adapting to what a caregiver can take up? The second is an evaluation question: how to tell whether the chatbot's moves land, and whether something therapeutically desirable follows in the caregiver's response. Answering it would take a finer-grained analysis than we offer here, one that considers the right level at which to read the exchange and what to attend to within it. This work contributes both empirical findings from field deployment and open questions for designing and evaluating LLM-based chatbots for caregivers' mental health responsibly.

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